Chapter 1 : Data preparation

Machine Learning Team
UP GL-BD

Learning outcomes

At the end of this module, the student will be able to:

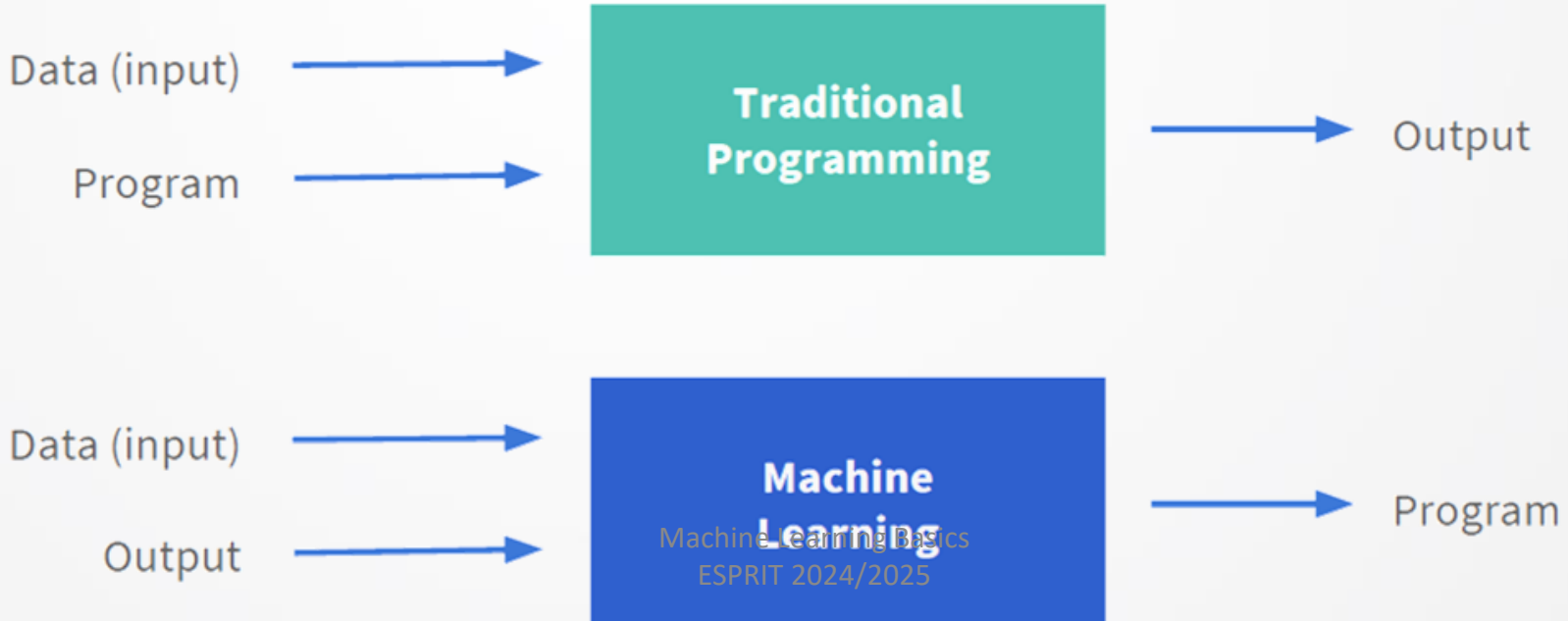
- Handle the various Python libraries intended for data science, namely Numpy, scipy, pandas, sklearn.
- Define the data preparation process:
  - Loading data as DataFrames
  - Data visualization
  - Data cleaning
  - Data mining
  - Data transformation
  - Data storage…
- Generate reduced-dimensional data using the “Principal Component Analysis” method

Introduction

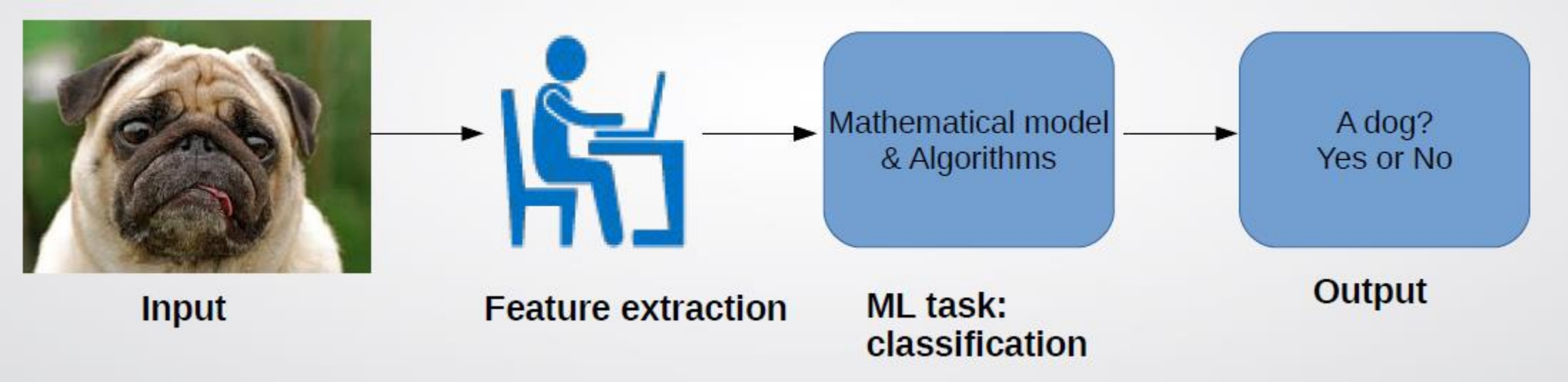
- Predictive modeling projects involve learning from data.
- The data refers to examples or cases from the field that characterizes the problem to be solved.
- On a Machine learning project, raw data typically cannot be used directly.
- This is because of reasons such as:
  - Machine learning algorithms require data to be numbers.
  - Some machine learning algorithms impose requirements on the data.
  - Statistical noise and errors in the data may need to be corrected…

What is Machine Learning?

- Machine learning is a subset of artificial intelligence.
- Machines have the ability to learn without being explicitly programmed but we need to provide a mathematical model, a database and a set of features.
- Computer program learns from examples with respect to a given mathematical model.



What is Machine Learning?

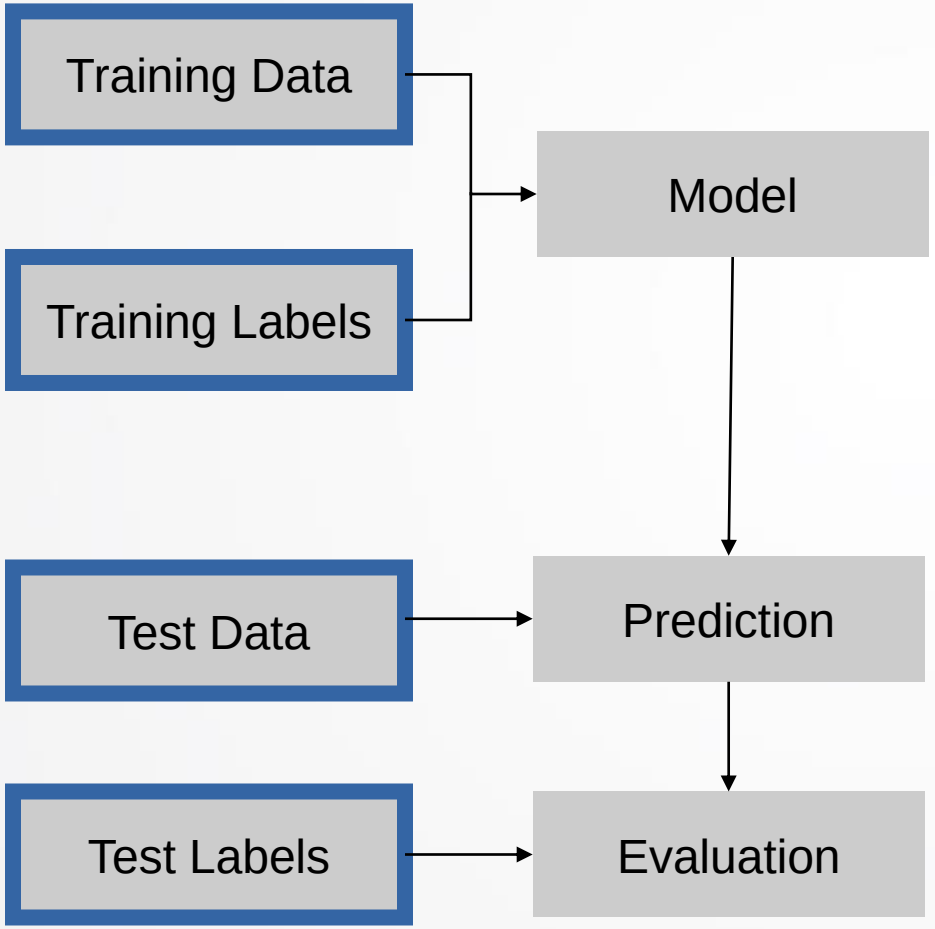


What is Machine Learning?

- **Supervised machine learning:**
  - learning from a labeled dataset.
  - Use mathematical model.
- **Unsupervised machine learning:**
  - Blind Analysis
  - Used when the dataset is not labeled.
  - Use mathematical models.
  - Use similarity measures.
- **Reinforcement machine learning:**
  - Used when there is no training dataset.

Machine Learning: How?

Supervised Learning



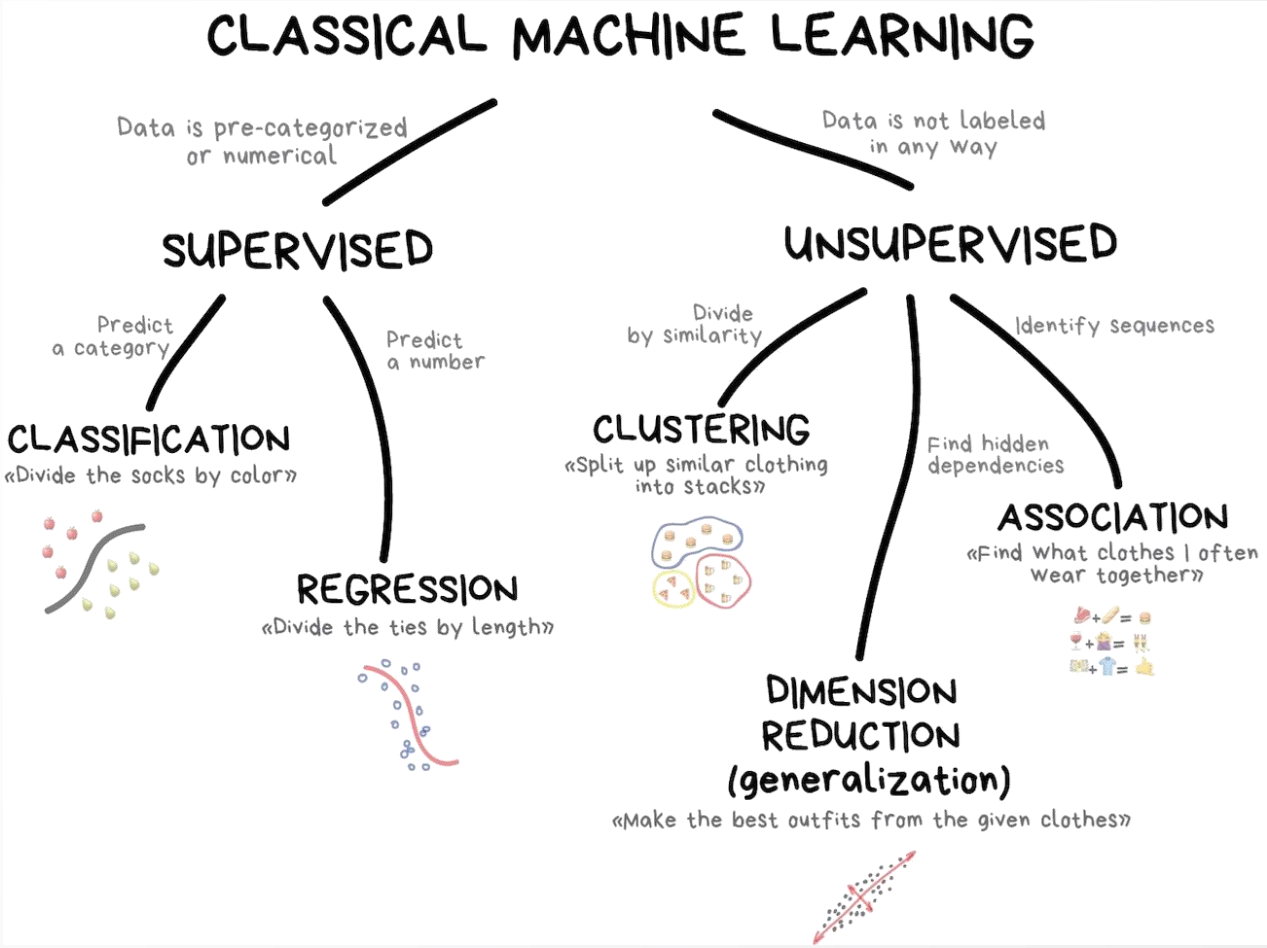
Subject	*t*	Feature 1	Feature 2	Target
Paul	1	1000	male	0
Paul	2	1100	male	0
Paul	3	1200	male	1
Paul	4	1300	male	1
Crista	4	20	female	0
Crista	5	100	female	0
Paulina	1	10000	female	0
Paulina	2	100000	female	1
Paulina	3	95000	female	1
Paulina	4	97000	female	1
Paulina	5	99000	female	1
Paulina	6	101000	female	1
George	1	50000	male	1
George	2	50000	male	1
George	3	50000	male	1
George	4	50000	male	1
George	5	50000	male	1
George	6	50000	male	1

Train

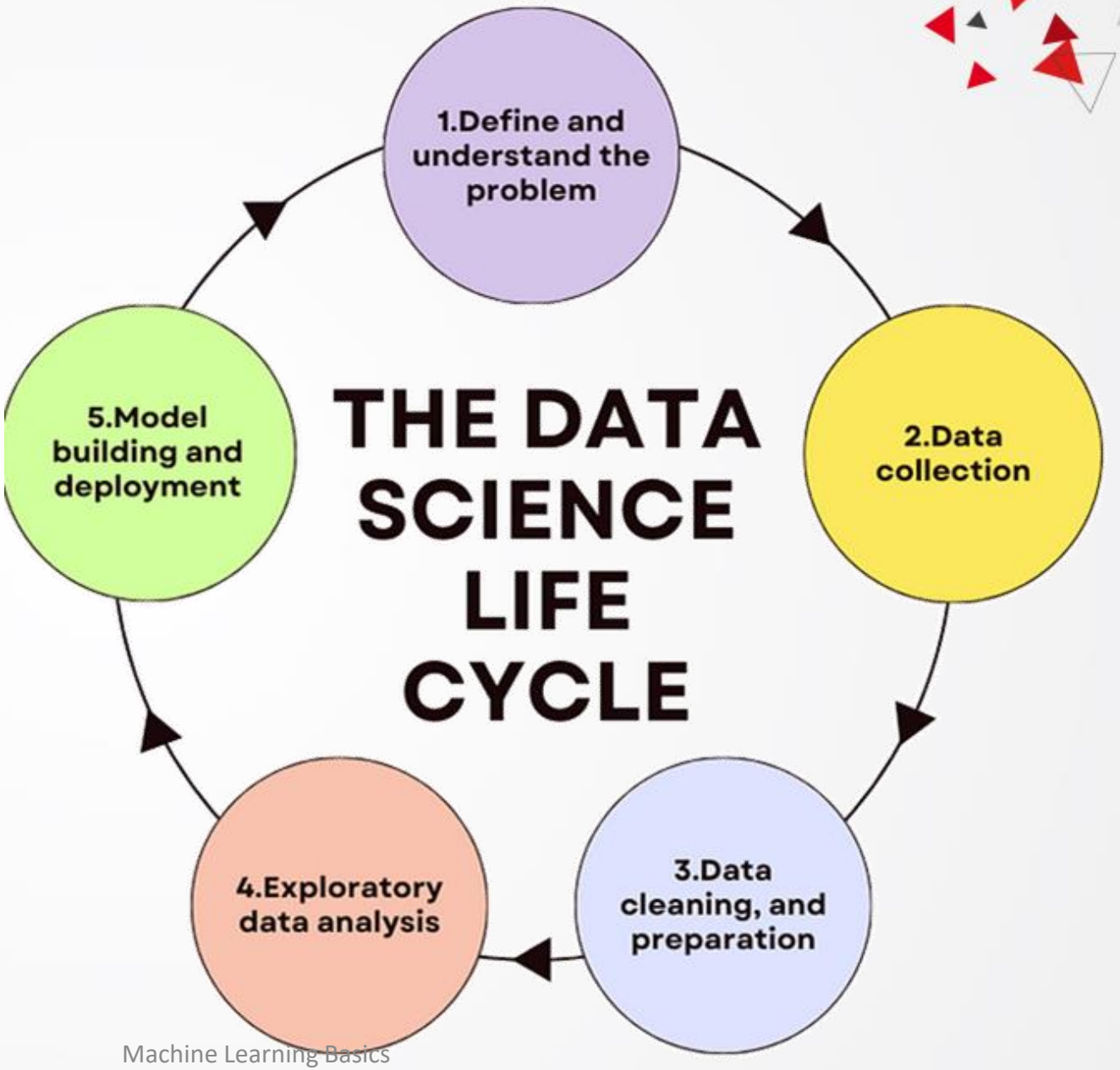
Test

Algorithmes en Machine learning

- **Supervised Learning models**
  - **Regression:** Used to predict a continuous value
  - **Classification:** Used to predict discrete value (a discrete class label output for an example)
- **Unsupervised Learning models**
  - **Clustering:** Based on similarity, grouping a set of objects the same group
  - **Dimension Reduction**
  - **Association rules**

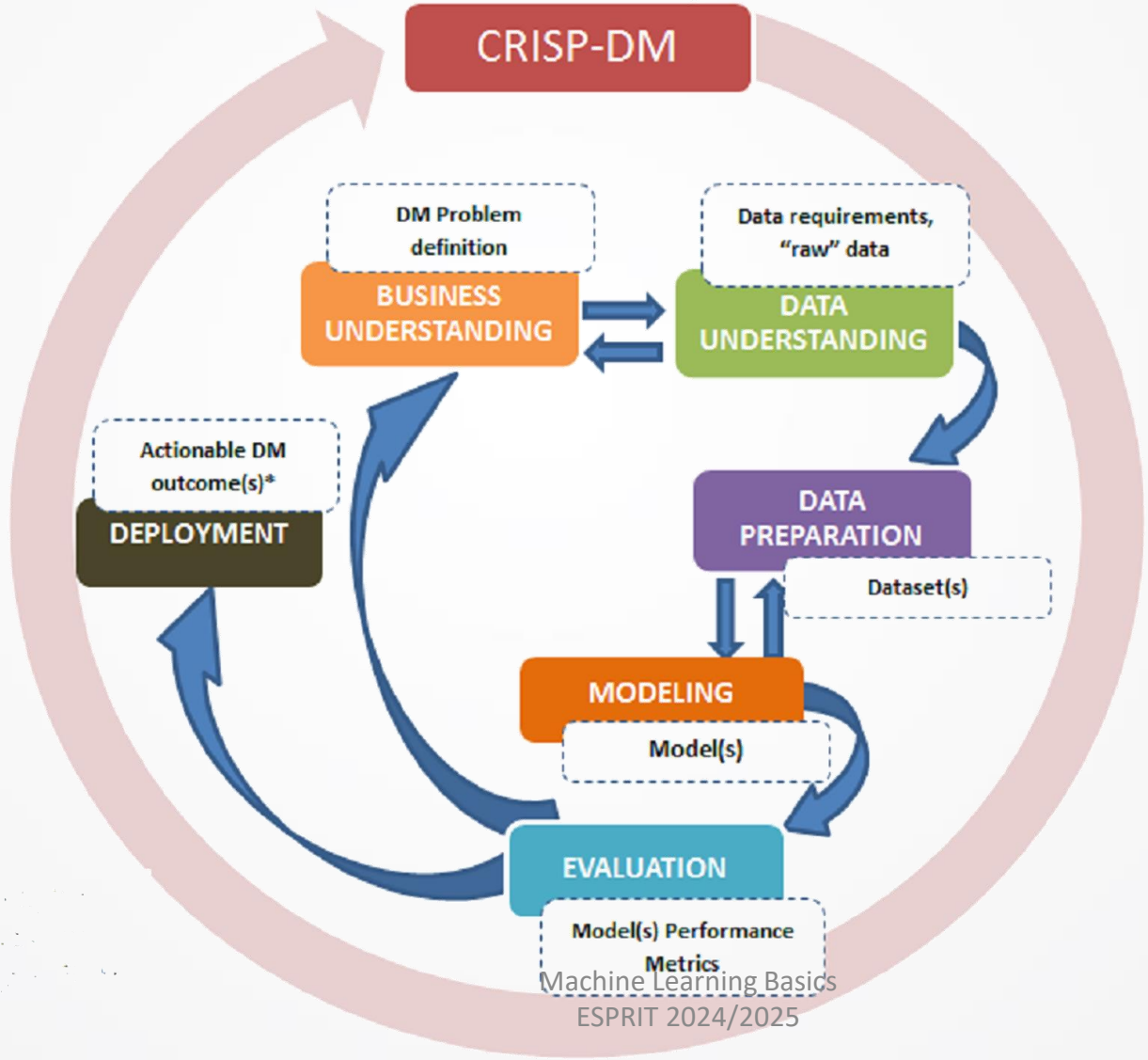


Data Science Life cycle

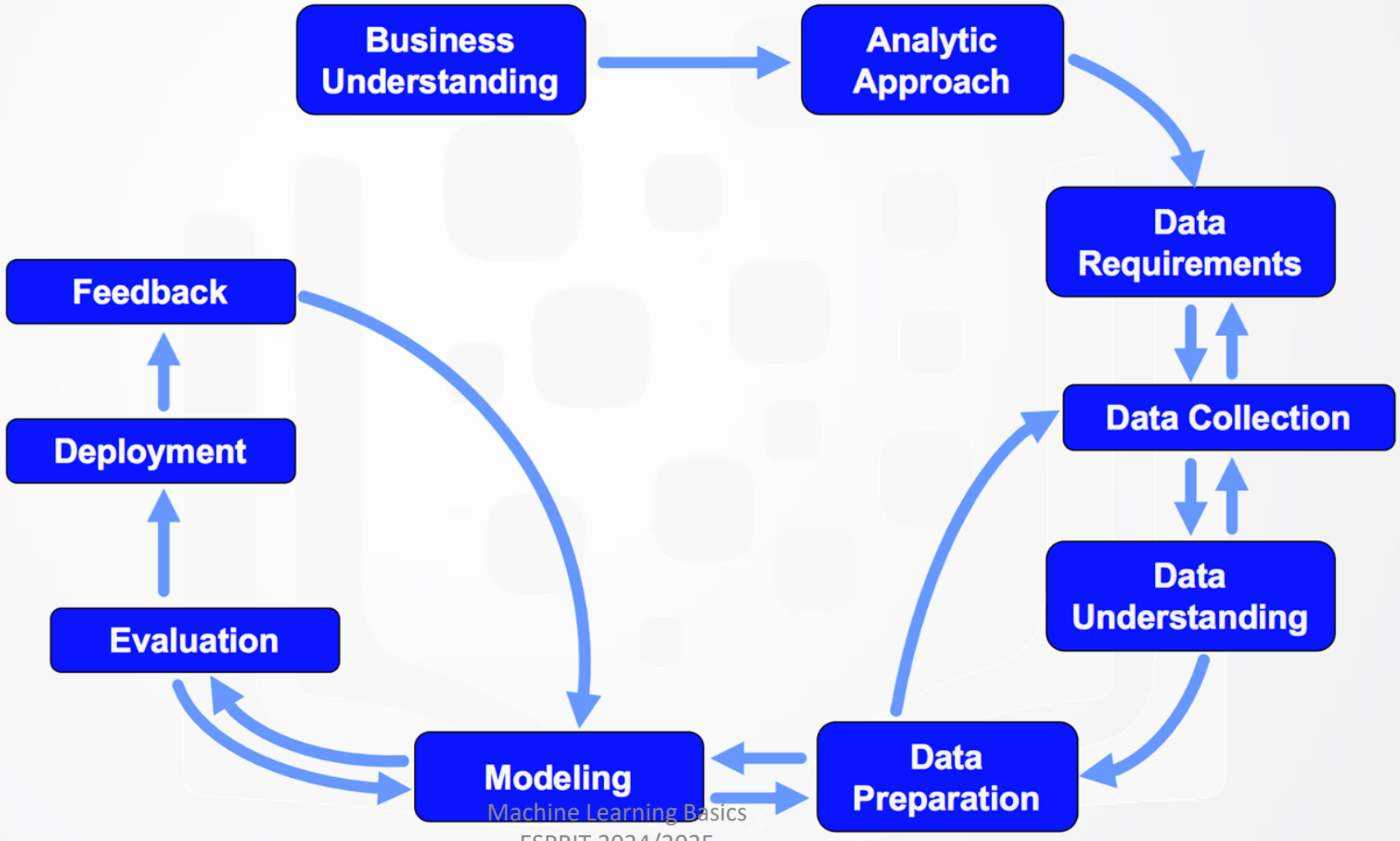


Data Science Driven Projects Methodologies

CRISP-DM methodology



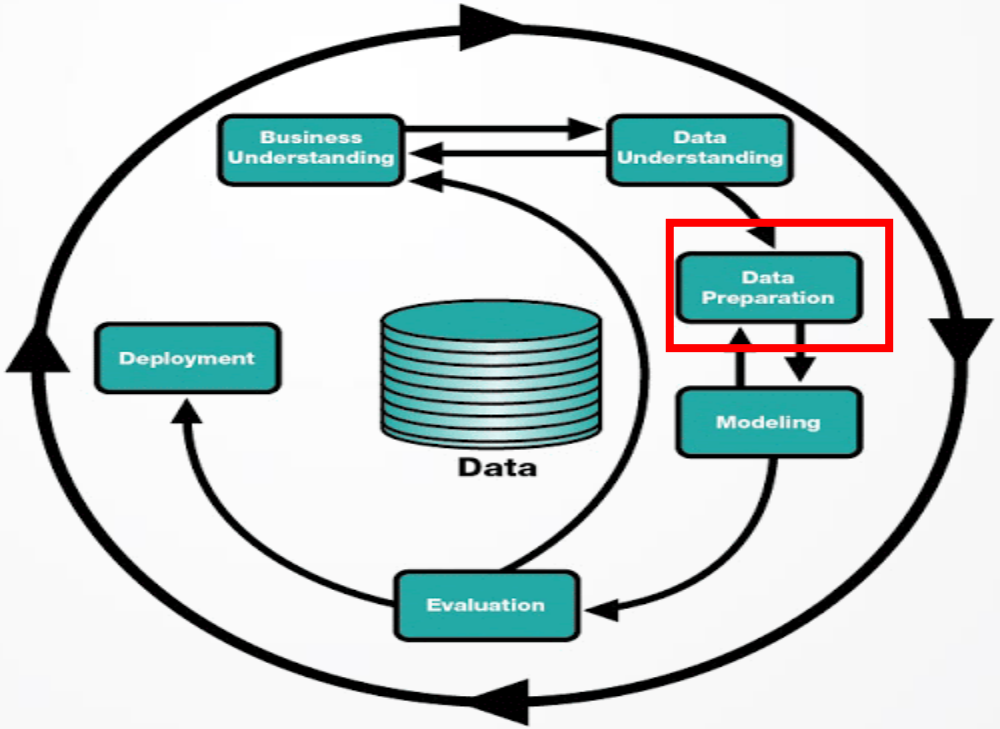
IBM Master Plan methodology



Microsoft TDSP methodology

Interest

- The raw data must be pre-processed prior to being used to fit and evaluate a machine learning model.
- This step in a predictive modeling project is referred to as **“data preparation“**



Data Preparation

Standard tasks of data preparation

Data Cleaning

Identifying and correcting mistakes or errors in the data.

Data Transforms

Changing the scale or distribution of variables.

Feature Engineering

Deriving new variables from available data.

Dimensionality Reduction

Feature Selection

Identifying those input variables that are most relevant to the task.

Principal component Analysis

Creating compact projections of the data.

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Principal component Analysis PCA

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Data Cleaning

"Data cleaning is used to refer to all kinds of tasks and activities to detect and repair errors in the data."

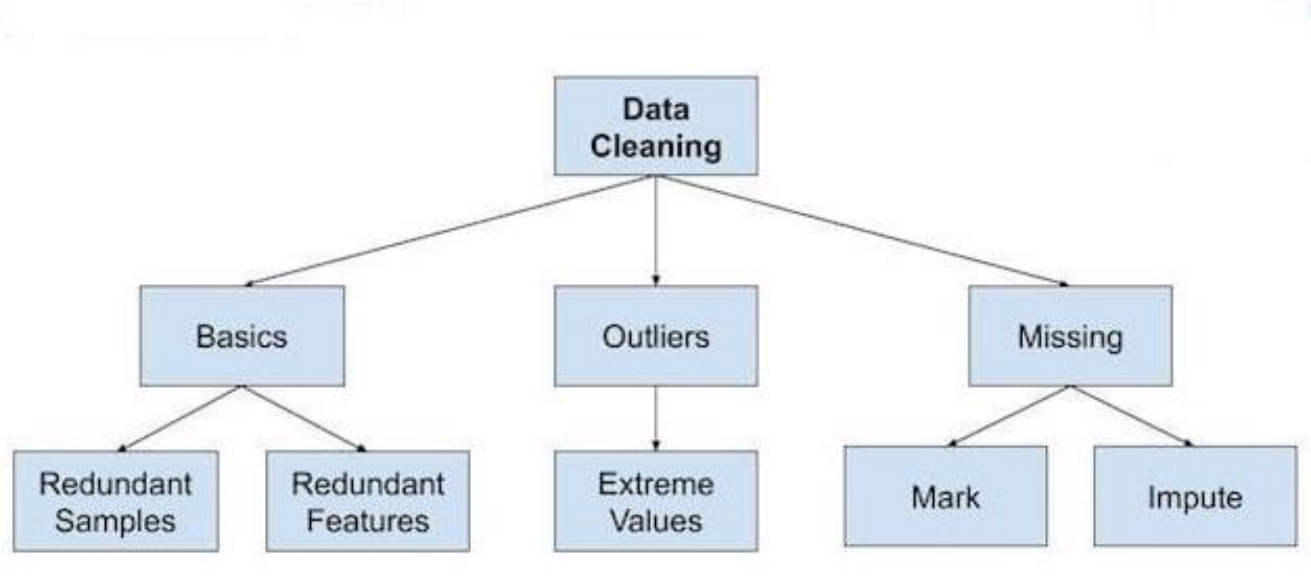
Page xiii, Data Cleaning, 2019.

"Cleaning up your data is not the most glamourous of tasks, but it's an essential part of data wrangling. […] Knowing how to properly clean and assemble your data will set you miles apart from others in your field."

Page 149, Data Wrangling with Python, 2016..

Data Cleaning

- Data cleaning includes simple tasks such as:
  - Define normal data.
  - Removing duplicate rows, redundant and irrelevant columns.
  - Identify outliers.
  - Dealing with missing values.



Data Cleaning: Redundant samples

pandas.DataFrame.duplicated

`DataFrame.duplicated(subset=None, keep='first')`

Return boolean Series denoting duplicate rows.

Considering certain columns is optional.

Parameters: **subset** : *column label or sequence of labels, optional*

Only consider certain columns for identifying duplicates, by default use all of the columns.

keep : *{'first', 'last', False}, default 'first'*

Determines which duplicates (if any) to mark.

- `first` : Mark duplicates as `True` except for the first occurrence.
- `last` : Mark duplicates as `True` except for the last occurrence.
- False : Mark all duplicates as `True`.

Returns: **Series**

Boolean series for each duplicated rows.

pandas.DataFrame.drop_duplicates

`DataFrame.drop_duplicates(subset=None, keep='first', inplace=False, ignore_index=False)`

Return DataFrame with duplicate rows removed.

Considering certain columns is optional. Indexes, including time indexes are ignored.

Parameters: **subset** : *column label or sequence of labels, optional*

Only consider certain columns for identifying duplicates, by default use all of the columns.

keep : *{'first', 'last', False}, default 'first'*

Determines which duplicates (if any) to keep. - `first` : Drop duplicates except for the first occurrence. - `last` : Drop duplicates except for the last occurrence. - False : Drop all duplicates.

inplace : *bool, default False*

Whether to drop duplicates in place or to return a copy.

ignore_index : *bool, default False*

If True, the resulting axis will be labeled 0, 1, …, n - 1.

Data Cleaning: Redundant features

```
# Quick examples:

# Drop duplicate columns
df2 = df.T.drop_duplicates().T


# Remove duplicate columns pandas DataFrame
df2 = df.loc[:,~df.columns.duplicated()]


# Use DataFrame.columns.duplicated() to drop duplicate columns
duplicate_cols = df.columns[df.columns.duplicated()]
df.drop(columns=duplicate_cols, inplace=True)
```

Data Cleaning: Outliers

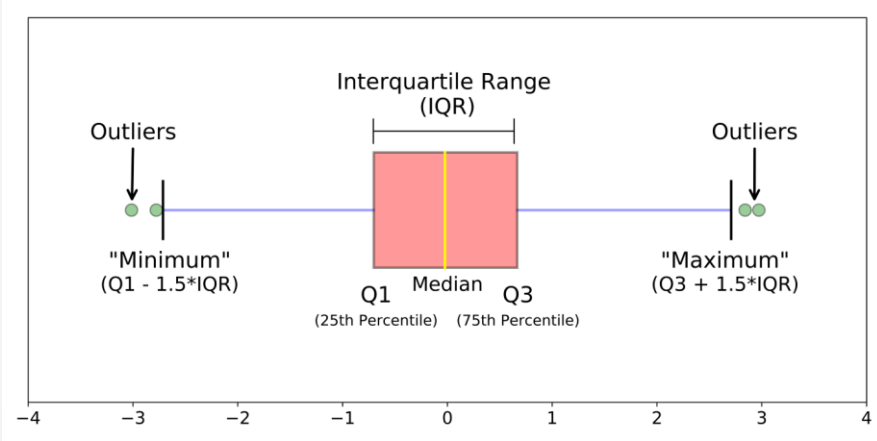
> *"We will generally define outliers as samples that are exceptionally far from the mainstream of the data."*
>
> *Page 33, Applied Predictive Modeling, 2013..*

Outliers causes:

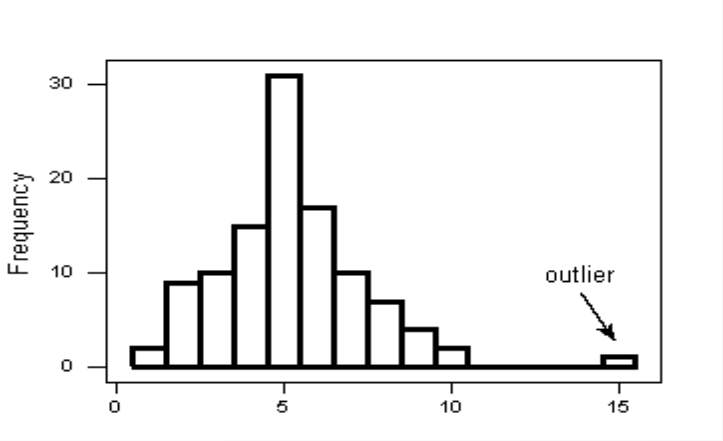
- Measurement or input error.
- Data corruption.
- True outlier observation…

Data Cleaning: Outliers

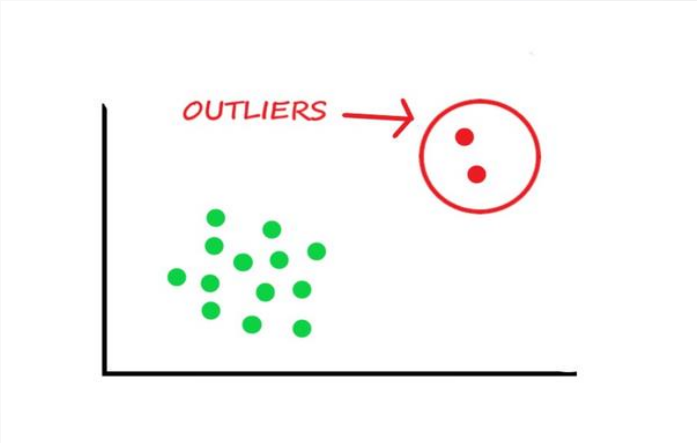
- Some methods to detect outliers:



Boxplot



Histogram



Scatter plot

Data Cleaning: Outliers

- Some methods to handle outliers:
  - Drop the outlier records.
  - Cap your outliers' data (min, max).
  - Assign a new value…

Data cleaning: Missing value

What is missing data?

- Missing data are defined as not available values, and that would be meaningful if observed.
- Missing data can be anything from missing sequence, incomplete feature, files missing, information incomplete, data entry error etc.

- Filling in missing values with data is called data imputation.

Data cleaning: Missing value

Some data imputation methods:

- Delete individuals with missing data
- Replace missing data with a fixed value
- Replace missing data with a decision tree
- Replace missing data with nearest values
- Replace missing data with dedicated algorithms…

Data cleaning: Missing value

A popular approach for data imputation is to calculate a statistical value for each column (such as a mean) and replace all missing values for that column with the statistic.

Strategy	« mean »	« median »	« most_frequent»	« constant»
Definition	Replace missing values using the mean along each column	Replace missing values using the median along each column	Replace the missing value using the most frequent value along each column	Replace missing values with fill_value. • fill_value=0 when imputing numeric data • fill_value=missing_value" for strings or object data types
Data type	numeric data	numeric data	numeric data/Strings	numeric data/Strings
Attention: Missing values must be marked with NaN				

Data cleaning: Missing value

	Python

« mean » strategy	**Méthode 1:**
	`sklearn.impute.SimpleImputer(*, missing_values=nan, strategy='mean', fill_value=None, verbose='deprecated', copy=True, add_indicator=False)`
	Méthode 2:
	`df.fillna(df.mean())`

«median» strategy	**Méthode 1:**
	`sklearn.impute.SimpleImputer(*, missing_values=nan, strategy='median', fill_value=None, verbose='deprecated', copy=True, add_indicator=False)`
	Méthode 2:
	`df.fillna(df.median())`

« most_frequent » strategy	**Méthode 1:**
	`sklearn.impute.SimpleImputer(*, missing_values=nan, strategy='most_frequent', fill_value=None, verbose='deprecated', copy=True, add_indicator=False)`
	
	Méthode 2:
	`df['columnName'] = df['columnName'].fillna(df['columnName'].mode()[0])`

« constant » strategy	`sklearn.impute.SimpleImputer(*, missing_values=nan, strategy='constant', fill_value=None, verbose='deprecated', copy=True, add_indicator=False)`

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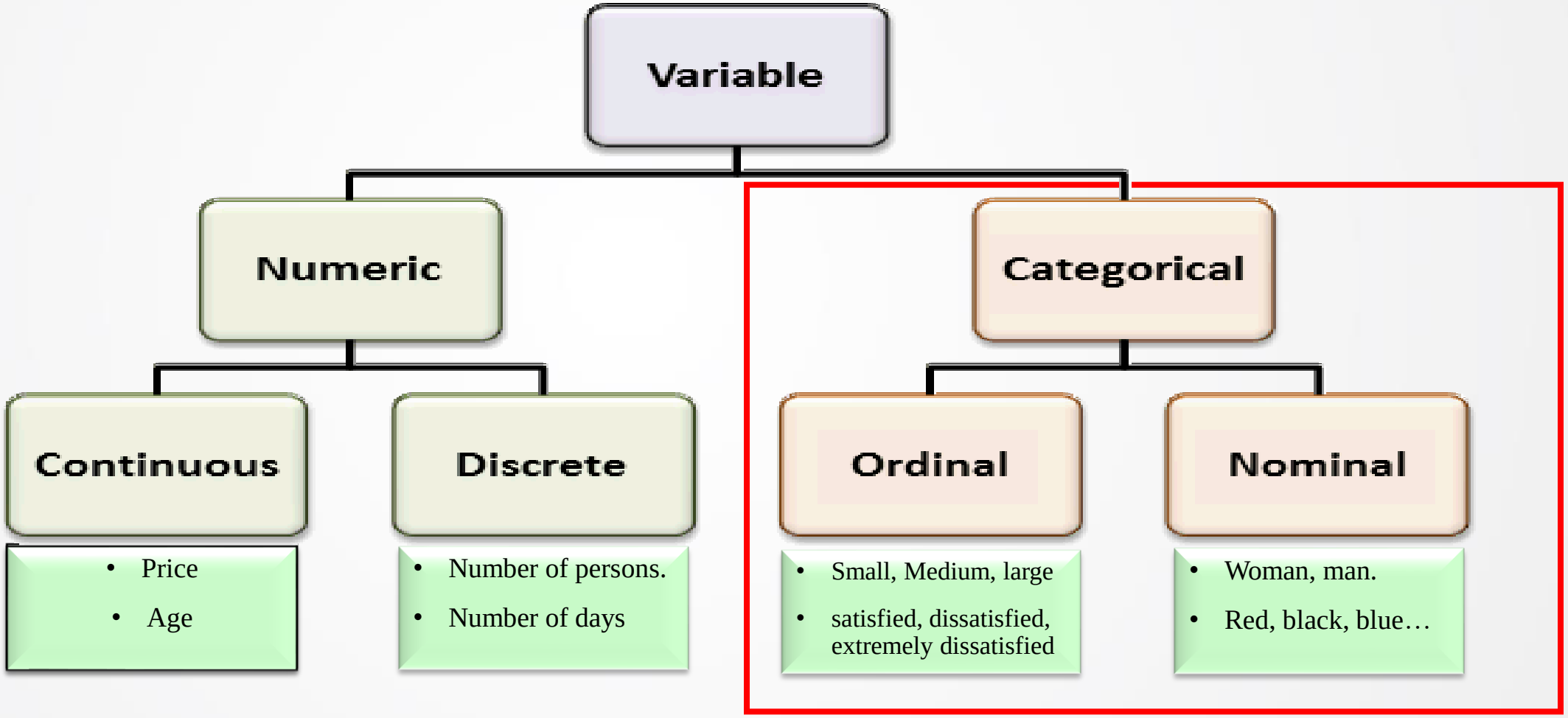
Principal component Analysis

Creating compact projections of the data.

Data Transformation

- Data transformation: is used to remove noise from a dataset.
- Noise is any distorted and meaningless data in a data set.

Data Transformation: Categorical features (1/3)



Data Transformation: Categorical features (2/3)

- Transforming categorical data is an essential step during data preprocessing. sklearn's machine learning library require the input dataset to always have numeric values it does not support categorical data.
- It is necessary to **convert** categorical features to a numerical representation.
- Before you start transforming your data, it is important to figure out if the feature you're working on is ordinal (as opposed to nominal). An ordinal feature is best described as a feature with ordered categories.

Data Transformation: Categorical features (3/3)

- Once you know what type of categorical data you're working on, you can pick a suiting transformation tool. In **sklearn** that will be:
  - A OrdinalEncoder or LabelEncoder for ordinal data,

Ordinal Encoding

workclass
State-gov
Self-emp-not-inc
Private
Private
Private

workclass
0
1
2
2
2

original dataset

x_1	x_2	y
5	8	calabar
9	3	uyo
8	6	owerri
0	5	uyo
2	3	calabar
0	8	calabar
1	8	owerri

LabelEncoder

```
{
  "calabar" ---> 0
  "owerri"  ---> 1
  "uyo      ---> 2
}
```

dataset with encoded labels

x_1	x_2	y
5	8	0
9	3	2
8	6	1
0	5	2
2	3	0
0	8	0
1	8	1

  - A OneHotEncoder for nominal data.

OneHot Encoding

workclass
State-gov
Self-emp-not-inc
Private
Private
Private

State-gov	Self-emp-not-inc	Private
1	0	0
0	1	0
0	0	1
0	0	1
0	0	1

Data transformation: Feature scaling

- Feature Scaling is a technique to standardize the independent features present in the data in a fixed range.
- It is performed during the data pre-processing to handle highly varying magnitudes or values or units.
- If feature scaling is not done, then a machine learning algorithm tends to weigh greater values, higher and consider smaller values as the lower values, regardless of the unit of the values.

Data transformation: Feature scaling

Standardization

Standardization:

$$z = \frac{x-\mu}{\sigma}$$

with mean:

$$\mu = \frac{1}{N}\sum_{i=1}^{N}(x_i)$$

and standard deviation:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2}$$

Data Transformation: Feature scaling

Normalization

- Some normalization methods are :
  - Maximum Absolute Scaling

$$x_{scaled} = \frac{x}{max(|x|)}$$

  - Min-max normalization

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

  - Decimal scaling
  - Z-score normalization…

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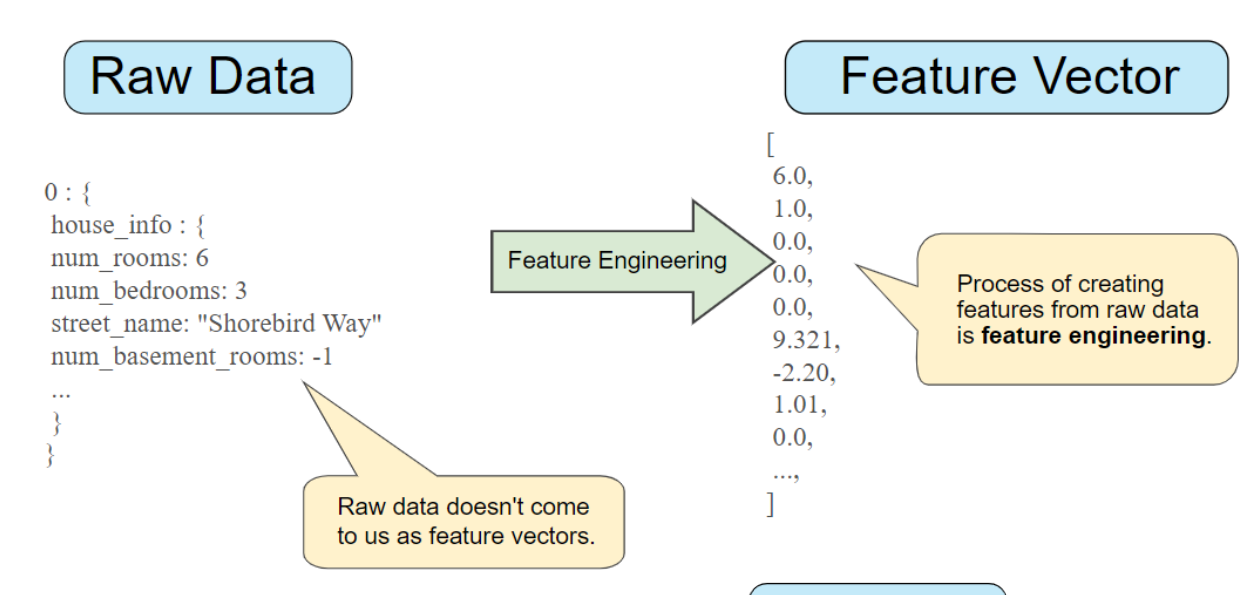
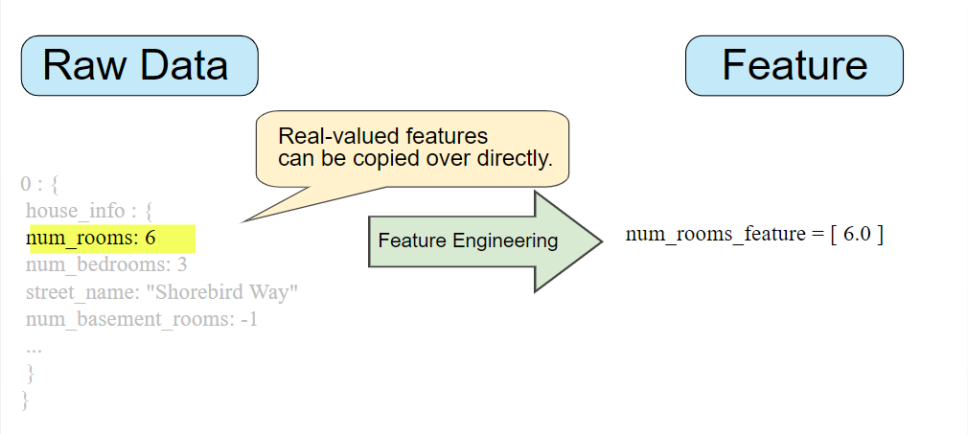
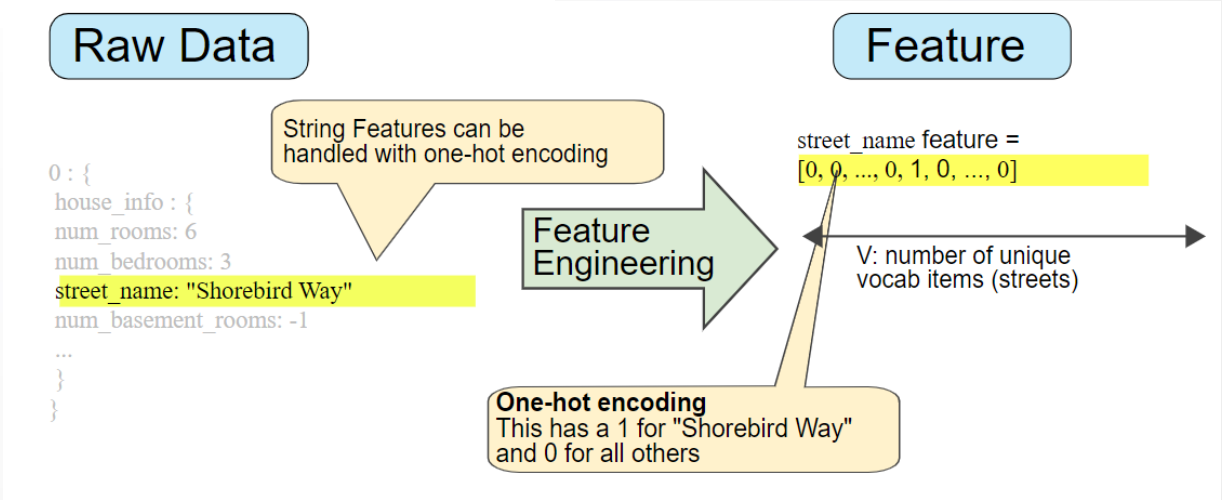
Feature engineering (1/2)

- Feature engineering is the process of transforming raw data into features that better represent the underlying problem to the predictive models, resulting in improved model accuracy on unseen data
- Example of Feature Engineering techniques:
  - **Creation of Features:** Sum, subtraction, average, min, max, product, quotient of the group of features.
  - **Extracting Features from a text**
  - **Topic extraction:** extract main topics from a text…
  - **Extracting features from an image**

Feature engineering (2/2)







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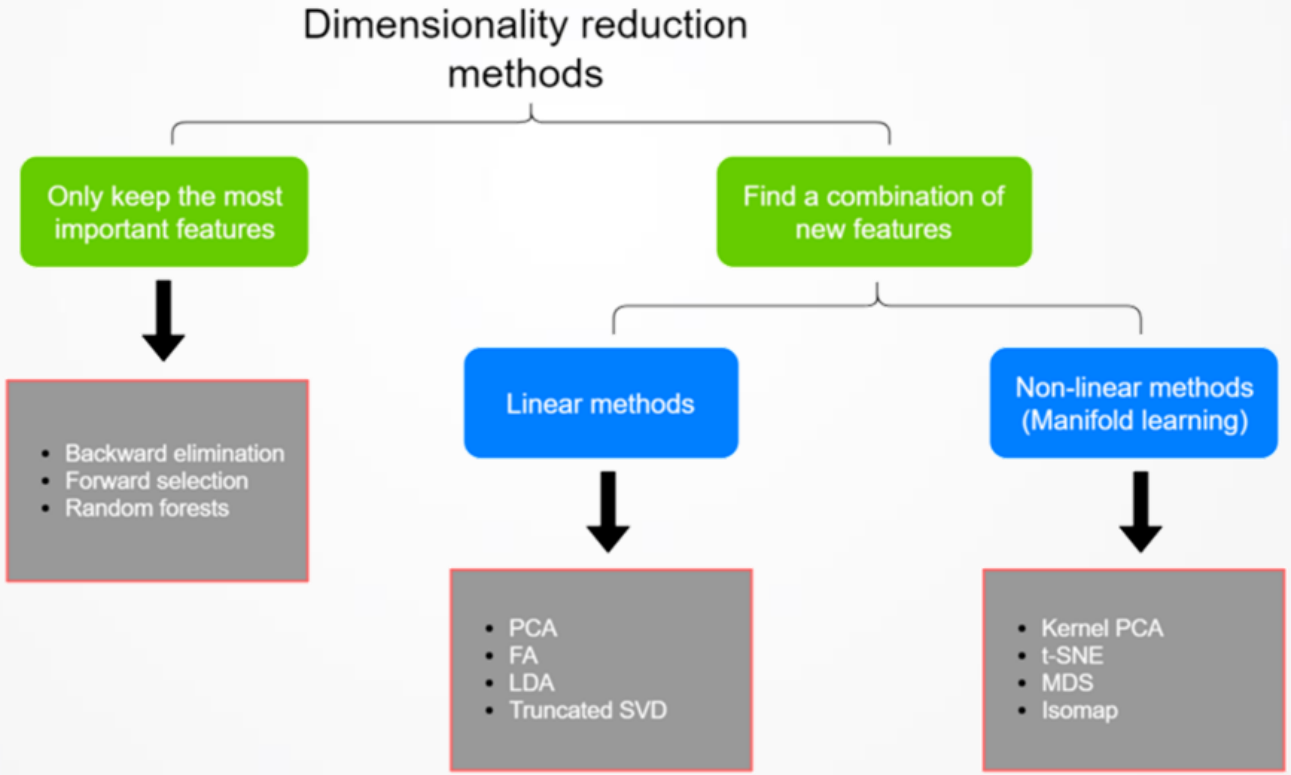
Dimensionality Reduction

- **Dimensionality reduction**, or **dimension reduction**, is the transformation of data from a high-dimensional space into a low-dimensional space so that the low-dimensional representation retains some meaningful properties of the original data.



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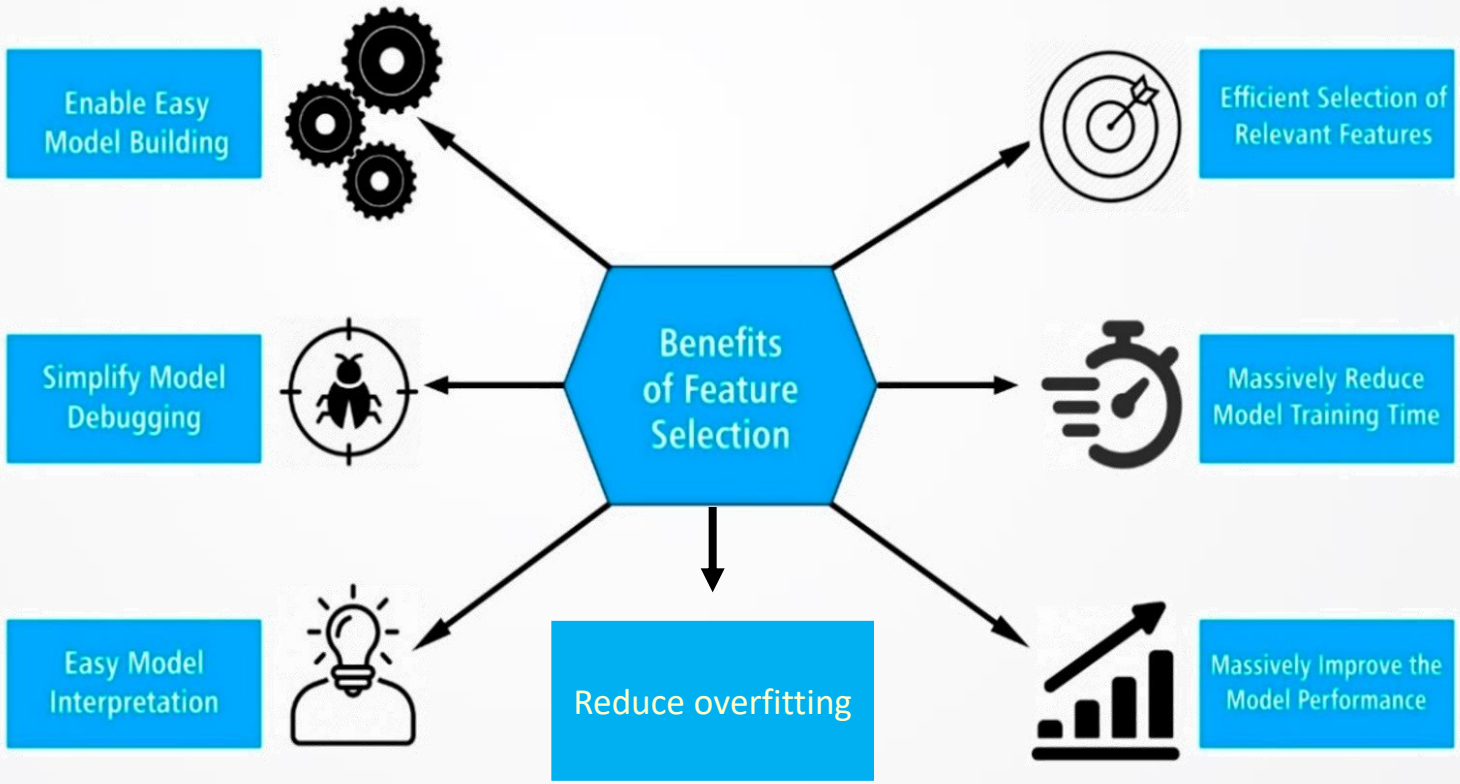
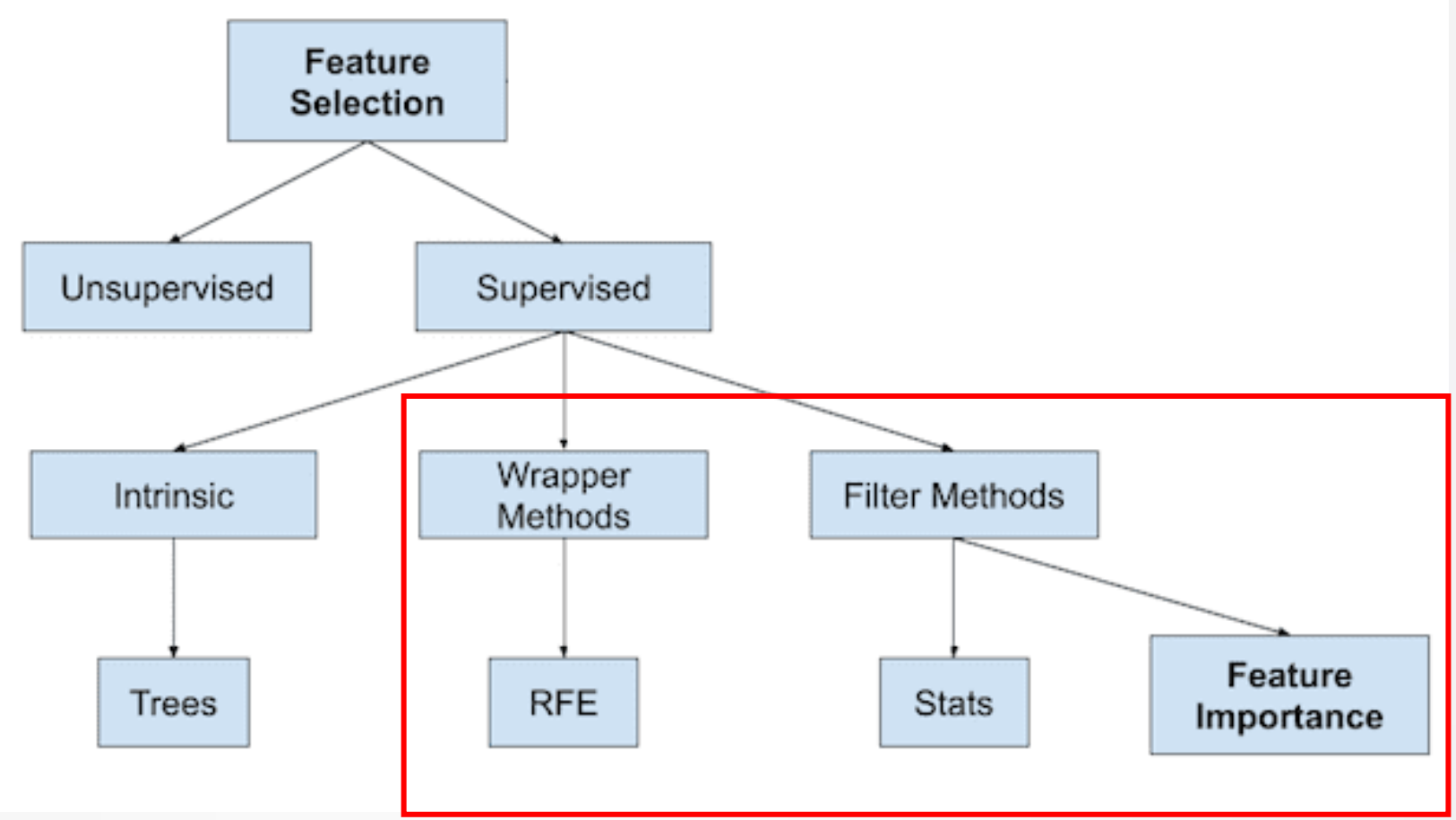
Feature Selection

- Feature Selection: is the process of selecting the most important features to input in machine learning algorithms

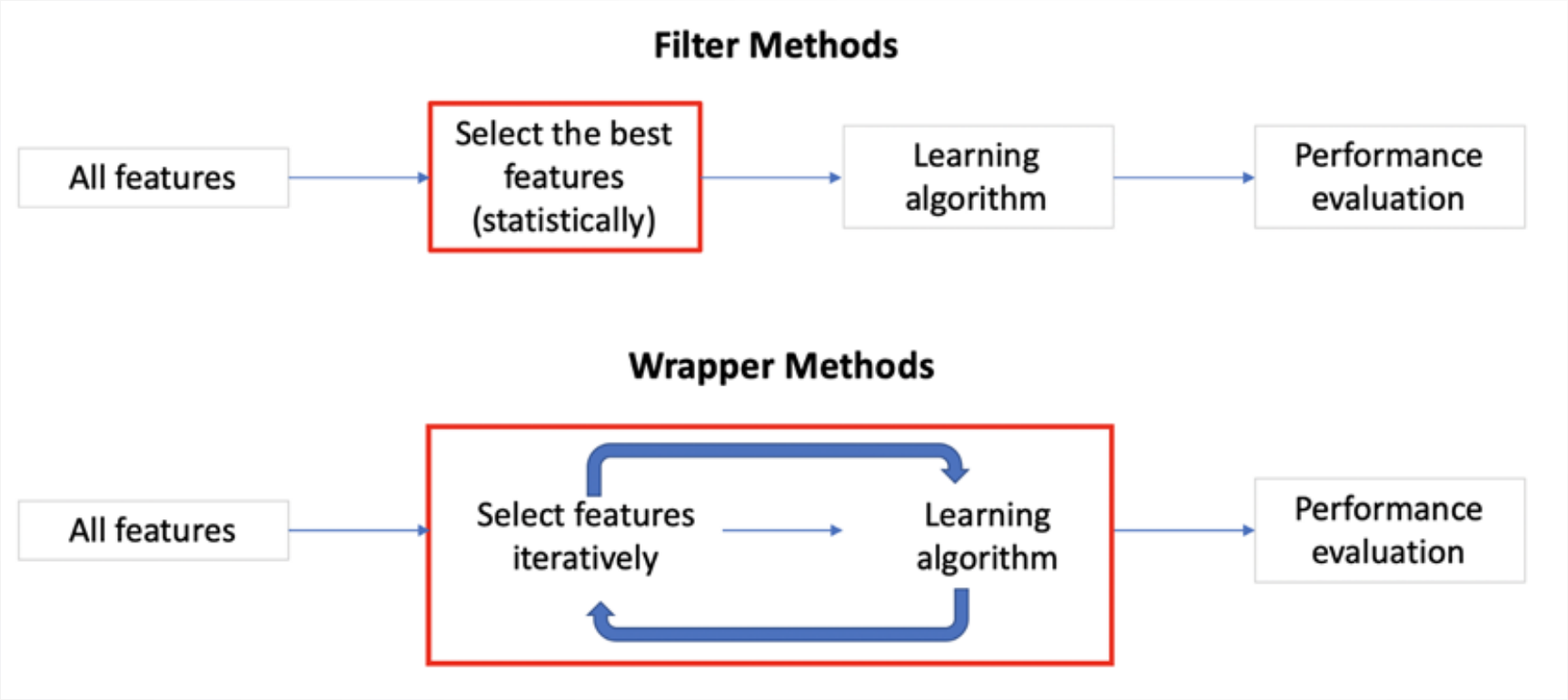


Feature Selection : Methods



Feature Selection : Methods

- **« Filter » vs. « Wrapper »:**



- Wrapper: choose in an iterative way the characteristics that give the best performing model.

- Filter: Assign a score to each input feature to select the best performing features.

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Dimensionality reduction advantages

- A lower number of dimensions in data means less training time and less computational resources and increases the overall performance of machine learning algorithms
- Dimensionality reduction avoids the problem of *overfitting*
- Dimensionality reduction is extremely useful for data visualization
- Dimensionality reduction removes noise in the data
- Dimensionality reduction can be used for image compression
- Dimensionality reduction can be used to transform non-linear data into a linearly-separable form…

Dimensionality Reduction: PCA

- **Principal component analysis (PCA)** is a statistical procedure that is used to reduce the dimensionality.
- PCA uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components.
- PCA is often used as a dimensionality reduction technique.

How PCA works?

PCA works on a process called **Eigenvalue Decomposition** of a covariance matrix of a data set.

The steps are as follows:

- First, calculate the covariance matrix of a data set.
- Then, calculate the eigenvectors of the covariance matrix.
  - The eigenvector having the highest eigenvalue represents the direction in which there is the highest variance. So this will help in identifying the first principal component.
  - The eigenvector having the next highest eigenvalue represents the direction in which data has the highest remaining variance and also orthogonal to the first direction. So, this helps in identifying the second principal component.
- Like this, identify the top ‘k’ eigenvectors having top ‘k’ eigenvalues to get the ‘k’ best principal components.

PCA steps

- Step 1: Standardize the Dataset
- Step 2: Find the Eigenvalues and eigenvectors
- Step 3: Arrange Eigenvalues
- Step 5: Transform Original Dataset

Example: PCA (1/11)

- Consider the following dataset :

x1	2.5	0.5	2.2	1.9	3.1	2.3	2.0	1.0	1.5	1.1
x2	2.4	0.7	2.9	2.2	3.0	2.7	1.6	1.1	1.6	0.9

- **Step 1: Standardize the Dataset**

Mean for x_1= 1.81 = x_{1mean}

Mean for x_2= 1.91 = x_{2mean}

We will change the dataset.

$x_{1new} = x_1 - x_{1mean}$	0.69	-1.31	0.39	0.09	1.29	0.49	0.19	-0.81	-0.31	-0.71
$x_{2new} = x_2 - x_{2mean}$	0.49	-1.21	0.99	0.29	1.09	0.79	-0.31	-0.81	-0.31	-1.01

Example: PCA (2/11)

- **Step 2: Find the Eigenvalues and eigenvectors**

Correlation Matrix c = $C = \left(\frac{X \cdot X^T}{N-1}\right)$

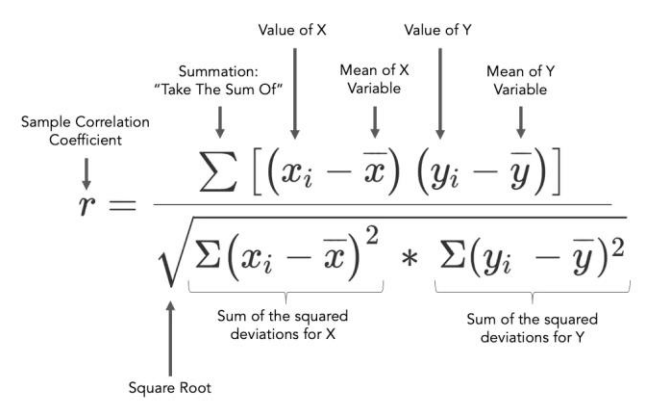
$$\begin{array}{cc} & \begin{array}{cc} \text{feature x1} & \text{feature x2} \end{array} \\ \begin{array}{c} \text{feature x1} \\ \text{feature x2} \end{array} & \begin{bmatrix} \text{var(x1,x1)} & \text{cov(x2,x1)} \\ \text{cov(x1,x2)} & \text{var(x2,x2)} \end{bmatrix} \end{array}$$

where, X is the Dataset Matrix

X^T is the transpose of the X

N is the number of elements = 10

$$C = \begin{bmatrix} 0.616556 & 0.615444 \\ 0.615444 & 0.716556 \end{bmatrix}$$



Example: PCA (3/11)

- The equation:

$|C - \lambda \times I|$=0, where { λ is the eigenvalue and I is the Identity Matrix }

is used to identify eigen values λ_i.

So solving this equation:

$$\begin{bmatrix} 0.616556 & 0.615444 \\ 0.615444 & 0.716556 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 0.616556 - \lambda & 0.615444 \\ 0.615444 & 0.716556 - \lambda \end{vmatrix} = 0$$

Taking the determinant of the left side, we get

$$0.44180 - 0.616556\lambda - 0.716556\lambda + \lambda^2 - 0.37877 = 0$$

$$\lambda^2 - 1.33311\lambda + 0.06303 = 0$$

We get two values for λ, that are $\boldsymbol{\lambda_1} = \mathbf{1.28403}$ and $\boldsymbol{\lambda_2} = \mathbf{0.0490834}$

Example: PCA (4/11)

- Now we have to find the eigenvectors for the eigenvalues λ_1 **and** λ_2**.**
- To find the eigenvectors from the **<u>eigenvalues</u>**, we will use the following approach:
  - First, we will find the eigenvectors for the eigenvalue 1.28403 by using the equation $C \times X = \lambda \times X$

$$\begin{bmatrix} 0.616556 & 0.615444 \\ 0.615444 & 0.716556 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 1.28403 \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 0.616556x + 0.615444y \\ 0.615444x + 0.716556y \end{bmatrix} = \begin{bmatrix} 1.28403x \\ 1.28403y \end{bmatrix}$$

Attention:
Here X does not represent the data set X(slide 41);
It's just the name of the vector
(During the course session, we named this vector V)

Example: PCA (5/11)

- Solving the matrices we get:

$$0.616556x + 0.615444y = 1.28403x \; ; \; \mathbf{x = 0.922049\, y}$$

(x and y belongs to the matrix X) so if we put y = 1, x comes out to be 0.922049.

So now the updated X matrix will look like:

$$X = \begin{bmatrix} 0.922049 \\ 1 \end{bmatrix}$$

Example: PCA (6/11)

IMP: Till now we haven't reached to the eigenvectors, we have to a bit of modifications in the X matrix. They are as follows:

A. Find the square root of the sum of the squares of the element in X matrix i.e.

$$\sqrt{0.922049^2 + 1^2} = \sqrt{0.850174 + 1} = \sqrt{1.850174} = 1.3602$$

B. Now divide the elements of the X matrix by the number 1.3602 (just found that)

$$\begin{bmatrix} \frac{0.922049}{1.3602} \\ \\ \frac{1}{1.3602} \end{bmatrix} = \begin{bmatrix} 0.67787 \\ 0.73518 \end{bmatrix}$$

So now we found the eigenvectors for the eigenvector λ_1, they are 0.67787 and 0.73518

Example: PCA (7/11)

- Secondly, we will find the eigenvectors for the eigenvalue 0.0490834 by using the equation {Same approach as of previous step)

So we found the eigenvectors for the eigenvector λ_2, they are 0.735176 and 0.677873

Sum of Eigen values λ_1 + λ_2 =1,33 = Total Variance
{Majority of variance comes from λ_1 }

Example: PCA (8/11)

- **Step 3: Arrange Eigenvalues**
- λ_1 is the eigenvector with the highest eigenvalue is the first Principal Component of the dataset.
- So in this case:
  - Eigenvector of λ_1 is the first principal component.
  - Eigenvector of λ_2 is the second principal component.

Example: PCA (9/11)

- **Step 4: Form Feature Vector**

$$\begin{bmatrix} 0.677873 & 0.735179 \\ 0.735179 & -0.677879 \end{bmatrix}$$

This is the FEATURE VECTOR for Numerical

- First column is the eigenvector of λ_1 & second column is the eigenvector of λ_2

Example: PCA (10/11)

- **Step 5: Transform Original Dataset**
- Use the equation Z = X V

$$\begin{bmatrix} 0.69 & 0.49 \\ -1.31 & -1.21 \\ 0.39 & 0.99 \\ 0.09 & 0.29 \\ 1.29 & 1.09 \\ 0.49 & 0.79 \\ 0.19 & -0.31 \\ -0.81 & -0.81 \\ -0.31 & -0.31 \\ -0.71 & -1.01 \end{bmatrix} \cdot \begin{bmatrix} 0.677873 & 0.735179 \\ 0.735179 & -0.677879 \end{bmatrix} = \begin{bmatrix} 0.8297008 & 0.17511574 \\ -1.77758022 & -0.14285816 \\ 0.99219768 & -0.38437446 \\ 0.27421048 & -0.13041706 \\ 1.67580128 & 0.20949934 \\ 0.91294918 & -0.17528196 \\ -1.14457212 & -0.04641786 \\ -0.43804612 & -0.01776486 \\ -1.22382.62 & 0.16267464 \end{bmatrix}$$

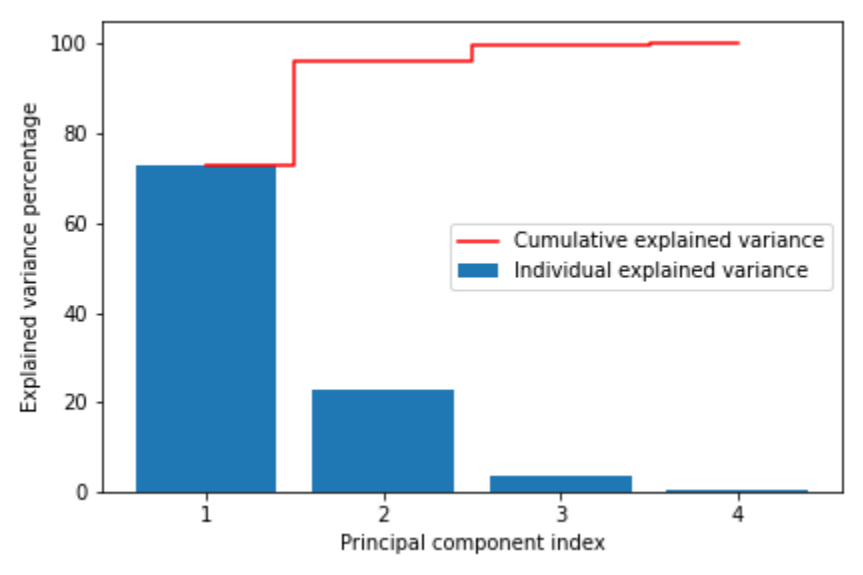
X V Z

Example: PCA (11/11)

- Organizing information into principal components in this way will allow you to reduce dimensionality without losing much information, by discarding components with little information and considering the remaining components as your new variables.

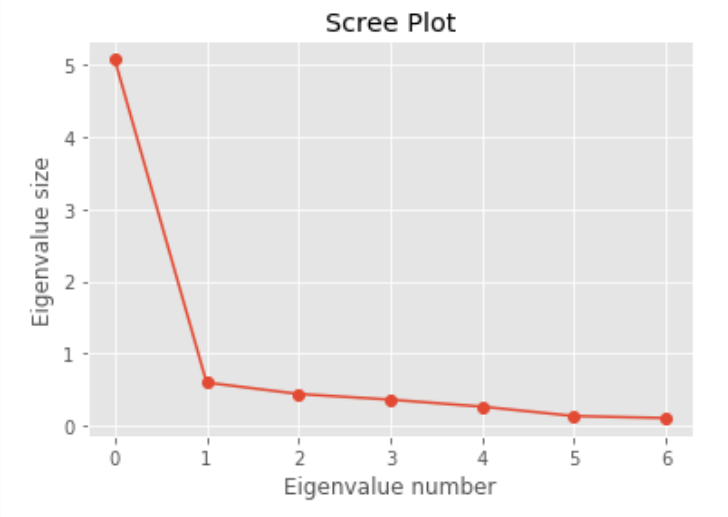
Best Number of Principal Components (1/2)

- **Method 1:** If your sole intention of doing PCA is for data visualization, you should select 2 or 3 principal components.
- **Method 2:** Plot the explained variance percentage of individual components and the percentage of total variance captured by all principal components.



Best Number of Principal Components (2/2)

- **Method 3:** Create the scree plot.

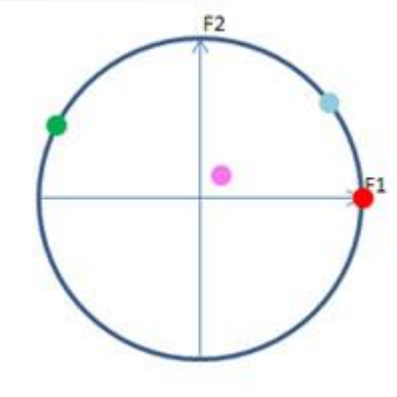


- **Method 4:** Follow the Kaiser's rule. According to the Kaiser's rule, it is recommended to keep all the components with eigenvalues greater than 1.
- **Method 5:** Use a performance evaluation metric such as RMSE (for Regression) or Accuracy Score (for Classification).

Interpretation: correlation circle

Understanding of variables

PCA results	Interpretation
Absolute variable position	Quality of the representation in the plan
Cosine angle between variables	Correlation



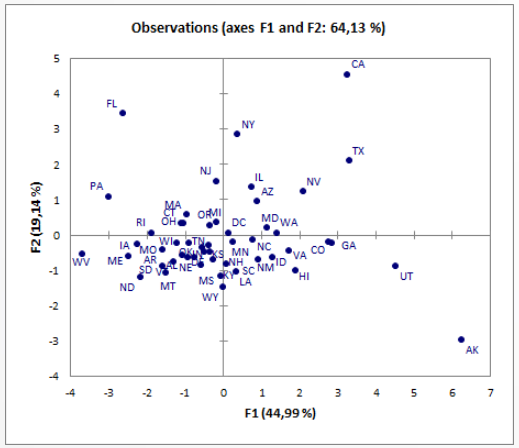
Correlation circle

- The closer the variables are to the edge of the circle, the better the variables are represented by the factorial plane, ie the variable is well correlated with the two factors constituting this plane.
- A point is said to be well represented on an axis or a factorial plane if it is close to its projection on the axis or the plane. If it is far away, it is said to be misrepresented.
- → angle formed between the point and its projection on the axis (the closer it is to 90 degrees, the less the point is well represented)

Interpretation: Individual factor map

Understanding the relationship between variables

PCA	Interpretation
Proximities between individuals	Similarities
Contribution of an individual to the α axis	Contribution to the inertia along the α axis



Individual factor map

The proximity in space between two well-represented individuals translates the resemblance of these two individuals from the point of view of the values taken by the variables. (and of course the difference otherwise)